2 A fixed mass of an ideal monatomic gas undergoes a cycle **ABCA** of changes, as shown in Fig. 2.1.

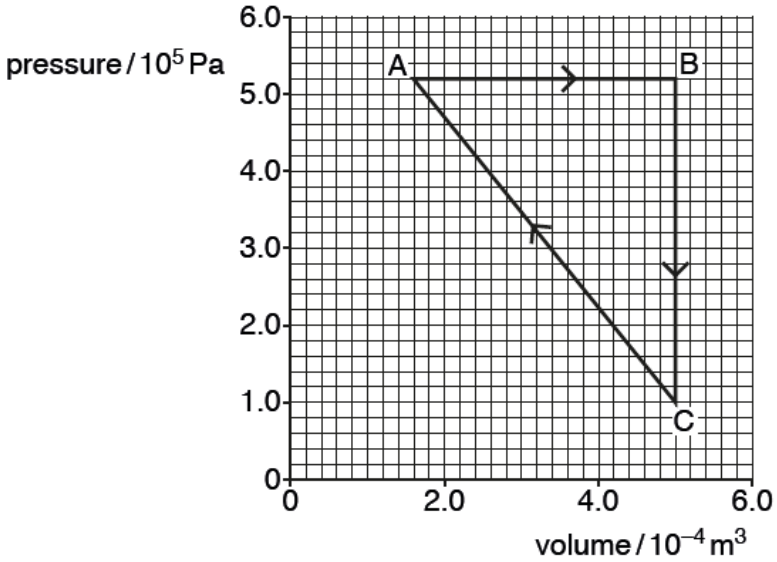

Fig. 2.1

(a) During the change from **B** to **C**, the internal energy of the gas decreases by 315 J.

By considering molecular energy, state and explain qualitatively the change, if any, in the temperature of the gas.

……..

……..

……..

…….. [2]

(b) During the change from **A** to **B**, the energy supplied to the gas by heating is 442 J.

Fig. 2.2 is a table of energy changes during one cycle. Complete Fig. 2.2.

section of cycle	heat supplied to gas / J	work done on gas / J	increase in internal energy of gas / J
A to B	442		
B to C			-315
C to A			

Fig. 2.2

[3]

Examiner's Use

(c) Calculate the root-mean-square speed of the gas atoms at point **B** if the root-mean-square speed at point **A** is 350 m s^{-1}.

root-mean-square speed = ……..……. m s^{-1} [2]